



Mobile phone screen defect detection based on improved YOLOv8n network[☆]

Peng Shi^a, Xueqin Li^{a,b,*}, Zhiming Feng^a, Xingwang Shang^a

^a Xihua University, 611730, Chengdu, China

^b Artificial Intelligence Key Laboratory of Sichuan Province, 643000, Yibin, China

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ABSTRACT

The continuous development of science and technology has led to mobile phones becoming an indispensable part of people's lives. The screens of used mobile phones have significant recycling value. However, the three most prevalent types of defects affecting screens—oil, scratches and stains—have a significant impact on the efficacy of their recovery. In light of the suboptimal precision of conventional defect detection techniques, this paper proposes a mobile phone screen defect detection method based on the enhanced YOLOv8n network. Additionally, a mobile phone screen defect detection device has been devised to facilitate the detection process. The enhanced model incorporates the CBAM attention mechanism within the SPPF module with a view to enhancing the multi-scale feature fusion capability of the model. In addition, a small target detection layer has been added with a view to enhancing the detection accuracy of defects that are small in size and contrast. Furthermore, the BiFPN module is integrated to enhance the performance of the feature pyramid network, facilitate the capture of multi-scale information. Finally, the issue of the lack of dynamic focusing of the Ciou loss function is addressed by replacing it with the WIoU. The experimental findings demonstrate that, in comparison with YOLOv8n, YOLOv8n-SDBW exhibits a mAP@50 enhancement of 0.83% and a mAP@50:95 improvement of 2.09%, while maintaining a minimal increase in model parameters and complexity.

1. Introduction

Advances in communications and networking technologies have led to the development of sophisticated electronic devices, resulting in the generation of large quantities of electronic waste. Waste electrical and electronic equipment (WEEE) has been identified as one of the most rapidly expanding waste streams globally, with an annual growth rate of 4%–5% [1]. As indicated by the Global E-waste Monitor 2020, the global volume of WEEE generated in 2019 reached 53.6 Mt, representing an increase of 9.2 Mt since 2014. Projections indicate that this figure will reach 74.7 Mt by 2030 [2]. However, only 15% of WEEE is formally treated globally [3–5], whereas there are still a plethora of reusable resources in used mobile phones [6]. It is therefore imperative that the recycling of used mobile phones be carried out.

Furthermore, the detection of defects in mobile phone screens represents a crucial aspect of the recycling process. However, the susceptibility of these screens to external factors, such as light, dust, and other environmental elements, poses a significant challenge. The traditional manual detection method is often inefficient, prone to errors, and subject to the influence of human subjectivity [7–9]. The defect detection method based on deep learning has been shown to be capable of compensating for the shortcomings of traditional detection methods.

However, the method requires a large amount of experimental data, and the detection accuracy for small targets is still insufficient. In order to address these issues, Brais Bosquet et al. have proposed a data enhancement method for small target detection [10]. This method combines a GAN-based object generator with object segmentation, image restoration and hybrid techniques. This approach has the potential to significantly reduce the resource consumption of data acquisition by deep learning methods, while enhancing the accuracy of small target detection.

In the context of surface defect detection, industrial cameras are typically selected as the image sensor for image acquisition [11]. Subsequently, traditional image processing methods [12] or deep learning methods [13,14] are employed for the extraction of features and the identification of defect types, including SSD [15], FasterRCNN [16], and YOLO [17,18]. For instance, in order to detect mobile phone cover glass defects, Yuan et al. proposed an automatic inspection system based on backlight scanning imaging technology and introduced a deep learning-based defect extraction method. The method has been shown to generate experimental data efficiently and to overcome the disadvantage of deep learning, which requires a large amount of data [19].

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* Corresponding author at: Xihua University, 611730, Chengdu, China.

E-mail addresses: freesia0408@163.com (P. Shi), 6326935@qq.com (X. Li), federer88@163.com (Z. Feng), 2646406336@qq.com (X. Shang).

In order to detect minor imperfections on steel surfaces, Xia et al. proposed an enhanced model, Bi-Level Efficient Global YOLO, which was based on YOLOv8x. The architecture of the model consists of a cross-stage partial DenseNet Global Feature Pyramid Network and a Bilayer Efficient Attention, which enhances feature fusion while supporting the sharing of information across different spatial scales and potentially non-adjacent semantic layers. According to the experimental results, the method is able to complete the defect detection task with high efficiency and high accuracy [20]. In order to detect defects in mobile phone screens, Zhu et al. proposed a Hierarchical Multi-Frequency based Channel Attention Net. To address the issue of handling various defect shapes, a deformable convolution-based ResNet network was introduced. Additionally, to mitigate the impact of extreme aspect ratios resulting from tiny phone surface defects, a RoI alignment technique was employed to reduce localisation errors [21].

A substantial corpus of research has demonstrated the efficacy and precision of deep learning-based defect detection methods, thus offering a solution to the shortcomings of traditional methods, namely their poor robustness and susceptibility to false detection. Nevertheless, many researchers concentrate on enhancing the accuracy of the detection methods, while neglecting the intricacies of parametric quantities and models. This oversight has resulted in suboptimal outcomes, including slow algorithmic reasoning, inadequate real-time performance, and substantial hardware requirements that fail to meet the standards for industrial defect detection. Therefore, the present paper investigates a deep learning-based defect detection system, with the aim of improving the detection accuracy of the model and achieving efficient defect detection.

2. Related work

2.1. Mobile phone screen defect detection

The recycling of mobile phone screens is significantly influenced by the presence and nature of defects, irrespective of whether the screen is defective or not. This paper examines three common defects observed in mobile phone screens: oil, scratches and stains. Figures (a), (b) and (c) in Fig. 1 illustrates these defects in detail. In order to detect these three types of defects, it is first necessary to employ traditional defect detection methods for the initial inspection of all mobile phone screens. During this process, larger size and higher contrast defects will be identified. However, the susceptibility of the detection results to environmental factors, such as light, renders them prone to defects with low contrast and some defects are of a relatively minor magnitude. These result in the traditional detection methods being often inaccurate and prone to leakage and misdiagnosis. This is demonstrated in Figures (d), (e) and (f). Accordingly, this paper puts forth an enhanced YOLOv8n network to address the shortcomings of the existing approach, namely, suboptimal efficiency and accuracy in the detection of mobile phone screen defects.

2.2. Introduction to YOLOv8n

The YOLOv8n algorithm is a rapid one-stage object detection method comprising an input segment, a backbone, a neck, and a head. The network structure diagram of YOLOv8n is shown in Fig. 2. The input segment performs mosaic data augmentation, adaptive anchor calculation, and adaptive greyscale padding on the input image. The backbone module is primarily responsible for feature extraction from the input image. This is achieved through the use of multiple Conv and C2f modules, which extract features from the image. The neck component is responsible for the multi-scale feature fusion of the feature map and the subsequent transfer of these features to the prediction layer. The neck component of YOLOv8n employs a FPN [22] + PAN [23] structure, which enhances the feature fusion ability of the model, while maintaining fast and efficient characteristics. The head part is responsible for the

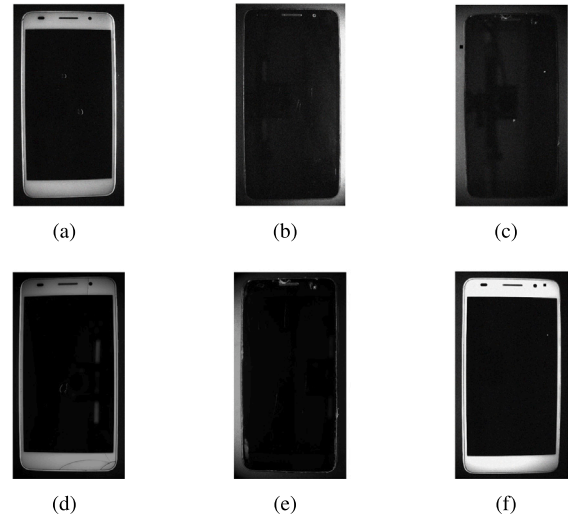


Fig. 1. Mobile phone screen defects; Figures (a), (b) and (c) show the normal defects in oil, scratches and stains; Figures (d), (e), and (f) show hard-to-detect oil, scratches and stains that are smaller in size and lower in contrast.

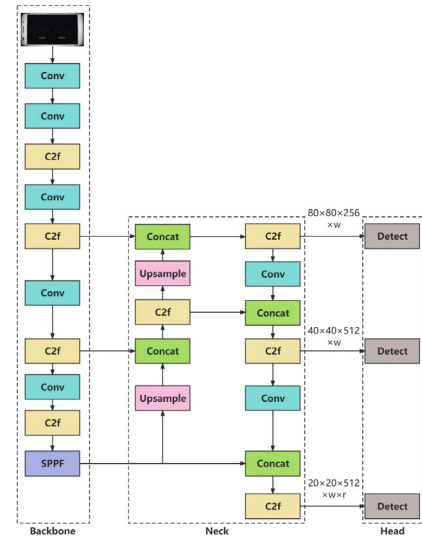


Fig. 2. YOLOv8 network structure.

final target detection or classification task, which employs a decoupled head and is divided into two branches: classification of classes (Cls) and regression of boxes (Box).

The YOLOv8n algorithm employs a lightweight network structure, which ensures rapid detection with high accuracy, and is well-suited for the detection of defects on mobile phone screens. However, the distribution of mobile phone screen defects is irregular, the contrast is low, and the size is often small. As a result, the existing YOLOv8n algorithm has low detection accuracy for these defects and a high leakage rate, which is insufficient to meet the requirements of mobile phone screen defect detection. It is therefore evident that in order to address the aforementioned issues, this paper proposes an enhancement to the model.

3. Proposed method

3.1. Improvement of the SPPF

In the YOLOv8 network, the SPPF module has been engineered to efficiently capture multi-scale information. As illustrated in Fig. 3(a),

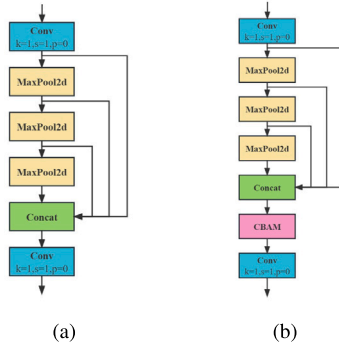


Fig. 3. A comparison is made between the SPPF and the SPPF_CBAM; (a) SPPF; (b) SPPF_CBAM.

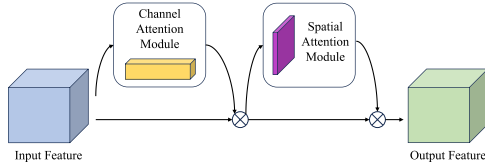


Fig. 4. The CBAM structure diagram.

the SPPF module utilises three maximum pooling layers to extract features at various scales. Subsequently, the three output feature maps undergo a splicing operation, thereby enabling the model to more accurately differentiate between different target objects and enhance the recall and accuracy of detection. However, the pooling process inevitably causes information loss, particularly with regard to the features of small-size targets, which become excessively blurred after multiple iterations of pooling. To address these issues, this paper proposes the incorporation of the CBAM attention mechanism [24] into the SPPF module to mitigate the aforementioned defects.

The structure of CBAM is illustrated in Fig. 4. The model consists of a channel attention module (CAM) and a spatial attention module (SAM) operating in tandem. The CAM module facilitates the capture of complex patterns and feature relationships in the data through pooling and fully connected network operations on different channels. This enhances the response of key feature channels and suppresses irrelevant or secondary channels, thereby enhancing the model's classification accuracy and generalisation ability. The SAM module learns the weights of different spatial locations of the feature map in dimension, highlights the location information of the target object in space, and helps the model to locate the target more accurately, so as to improve the accuracy and precision of the target frame. As demonstrated in Fig. 3(b), this paper proposes the integration of the CBAM attention mechanism into the SPPF module. This integration enables the adaptive adjustment of the feature map weights, enhancing the multi-scale feature fusion capability.

3.2. Improvement of the detection layer

The defects observed in mobile phone screens are characterised by a multitude of imperfections, including low resolution and restricted channel characteristics, such as minute scratches. In the process of identifying mobile phone screen defects, it has been noted that the recall value for tiny scratches in the resulting detection is relatively low. This suggests that a considerable number of such defects may have been overlooked. This is due to the fact that the downsampling multiplier of YOLOv8 is relatively high, which makes it challenging for the deeper feature map to effectively learn the feature information of small targets. Accordingly, this paper proposes the incorporation of a supplementary

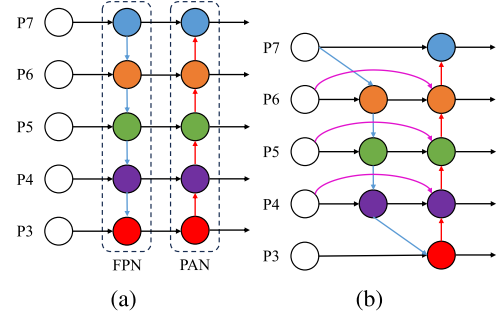


Fig. 5. A comparison is made between the FPN-PAN and the BiFPN; (a) FPN-PAN; (b) BiFPN.

small target detection layer, whereby the shallower feature maps are spliced with the deeper feature maps for detection purposes. This approach is designed to enhance the network's capacity to discern small targets, while maintaining a manageable computational load.

3.3. Improvement of the BiFPN

The BiFPN is a neural network architecture for target detection and semantic segmentation tasks that aims to enhance the performance of feature pyramid networks. The BiFPN was initially introduced by Google in the EfficientDet [25] object detection algorithm. The BiFPN introduces bidirectional connectivity, which allows information to propagate bi-directionally between different resolution levels. Furthermore, it employs an adaptive feature tuning mechanism to adjust the features at different levels in order to better match the needs of different tasks. This helps to improve feature fusion and thus enhances the network's ability to handle objects of different sizes and complexity. A comparison of the BiFPN method with the FPN-PAN approach employed by YOLOv8n reveals that the former possesses a superior capacity to extract multi-scale feature information. Fig. 5 depict the schematic structure of FPN-PAN and BiFPN. Therefore, in order to enhance the multi-scale information extraction capability and model detection accuracy, the BiFPN module is incorporated into Concat.

3.4. Improvement of the WIoU

The intersection and concatenation ratio IoU is generally employed in the target detection process to evaluate the similarity of the region between the prediction results and the actual annotation, i.e., the ratio of intersection and concatenation [26]. The YOLOv8n network model utilises the CIOU loss function to calculate the regression loss of the bounding box. However, due to the absence of dynamic focusing in the CIOU, which leads the inability of the CIOU to flexibly allocate the gradient gain according to the quality of the anchor frames. The WIoU v3 loss function [27] enables the bounding box regression to prioritise ordinary quality anchor frames by introducing a dynamic non-monotonic focusing mechanism. This mechanism serves to reduce the competitiveness of high-quality anchor frames while concomitantly reducing the harmful gradient generated by low-quality anchor frames. Consequently, this paper proposes the utilisation of the WIoU v3 loss function as a replacement for the CIOU loss function, with the objective of enhancing the detection accuracy of the model. The calculation of the WIoU v3 loss function is outlined in Eq. (1) through Eq. (5).

$$L_{WIoU\ v3} = r R_{IoU\ v3} L_{IoU} \quad (1)$$

$$r = \frac{\beta}{\delta (\alpha)^{\beta-\delta}} \quad (2)$$

$$R_{IoU\ v3} = \exp \left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(W_g^2 + H_g^2)^*} \right) \quad (3)$$

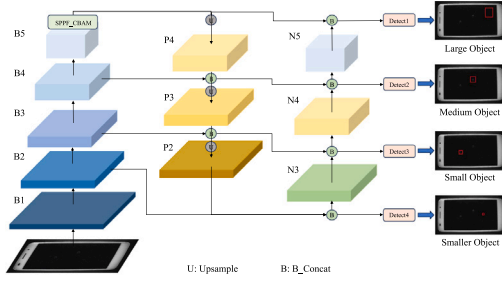


Fig. 6. YOLOv8n-SDBW network structure.



Fig. 7. Defect detection systems.

$$L_{IoU} = 1 - IoU \quad (4)$$

$$\beta = \frac{L_{IoU}^*}{L_{IoU}} \in [0, +\infty) \quad (5)$$

where r is the non-monotonic focusing coefficient; $R_{WIoU v3}$ is the penalty term; L_{IoU}^* is the monotonic focusing coefficient; L_{IoU} is the IoU loss; β is the degree of outlier; δ is the gradient gain; and α is the equilibrium factor, with the superscript indicating that a separating operation is performed.

3.5. Improved network structure

In this paper, the four aforementioned improvement methods are integrated into the YOLOv8n network, resulting in the creation of the improved network model, designated as YOLOv8n-SDBW. The letters “S”, “D”, “B” and “W” represent the following modifications: “S” is used to denote the integration of the CBAM module into the SPPF module, “D” stands for adding a small target detection layer, “B” stands for integrating the BiFPN module in the neck part, and “W” stands for replacing the CIoU loss function with WIoU v3.

The modified network structure is illustrated in Fig. 6. The YOLOv8n-SDBW network maintains the YOLOv8n backbone network, incorporating the CBAM module into the SPPF module (denoted SPPF_CBAM module) to enhance the algorithm’s detection accuracy, and integrating the BiFPN module into the Concat module (designated B_Concat module) to enhance the feature information capturing ability and generalisation capability of the algorithm. The addition of a small target detection layer with a feature map size of 160×160 is then proposed for detecting targets larger than 4×4 . Finally, CIoU is replaced with WIoU to improve the detection accuracy of the model.

4. Experiments

4.1. Defect detection systems

In order to complete the mobile phone screen defect detection efficiently, this paper proposes a system for the automatic detection of defects in mobile phone screens, as shown in Fig. 7. The system is composed of a conveyor belt, a defect detection device, a coaxial surface light source and a six-axis robot. The conveyor belt on the left side of the system is responsible for the transmission task of the mobile phone. When the mobile phone passes through the defect detection

device, the photoelectric sensor triggers the camera to take a picture. The picture is then pre-processed and defects are detected by using the improved algorithm proposed in this paper. Finally, the classification of defects is then carried out by the six-axis robot according to the type of defect.

4.2. Production of datasets

This paper addresses three categories of defects that are commonly observed on mobile phone screens and which are challenging to identify: oil, scratches and stains. These defects are characterised by their small size and low contrast, and the paper presents a comprehensive collection of 2,000 images for each category, resulting in a total of 6,000 images. The dataset is then divided into a training set, a validation set and a test set in a ratio of 6:2:2. Given the limited data set, a data augmentation method was employed to expand the dataset and enhance the model’s generalisation ability [28]. The data enhancement methods employed in this study comprise a random rotation, flipping, addition of noise, panning, cropping, adjustment of brightness, and application of masking techniques, which collectively expand the original training set of 3600 sheets to 10,800 sheets. This effectively enhances the model’s generalisation ability and robustness. Finally, the image data underwent labelling using the Labellmg software.

4.3. Evaluations metrics

The evaluation metrics employed in this study for target detection algorithms are recall, mAP (mean Average Precision), FPS (Frames Per Second), params (Parameters) and GFLOPs (Giga Floating-point Operations Per Second).

Recall: The recall is used to indicate the probability that the model will correctly predict a positive sample if it is, in fact, a positive sample. It reflects the capacity of the classifier to identify all true positive samples and is typically employed to evaluate the potential for model leakage. The formula is provided in Eq. (6).

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

In the formula, TP signifies that the sample prediction is positive and accurate, whereas FN denotes that the sample prediction is negative but erroneous.

mAP: In the context of multi-category target detection, mAP represents the mean of the AP values for each category. The calculation formula is provided in Eq. (7). mAP@50 represents the average of the APs for each category when the confidence level (IoU) is set to 50%. mAP@50:95 denotes the interval of 5% from the mAP threshold of 50% to the mAP threshold of 95%. Ten mAP values are obtained, and then the average is taken for these ten values.

$$mAP = \frac{\sum AP}{N} \quad (7)$$

In the formula, N represents the number of categories. The AP is employed to denote the area under the Precision–Recall curve.

FPS: In computing, FPS refers to the number of images that can be processed in a given time period, typically expressed in units of seconds. This value represents the model’s performance in processing images, with a higher FPS indicating a faster detection rate.

Parameters: The parameters is used to describe the size of the model. Smaller models can be deployed to mobile devices with limited resources with greater ease, which in turn reduces the overheads associated with downloading and storing the model.

GFLOPs: GFLOPs is a unit used to quantify the computational performance of a model during inference or training, as well as the complexity of the model itself.

4.4. Ablation experiment

The results of the ablation experiments are presented in Table 1. The detection results of the YOLOv8n network demonstrate that it exhibits the lowest detection accuracy for Scr defects (92.77%)

Table 1
Results of ablation experiment.

Model	P/%			R/%	mAP@50/%	mAP@50:95/%	Params/M	FLOPs/G	FPS
	Oil	Scr	Sta						
YOLOv8n	99.28	92.77	98.73	94.04	96.87	67.04	3.01	8.1	113.64
YOLOv8n-S	98.46	94.24	98.86	94.11	97.00	68.04	3.27	8.3	112.36
YOLOv8n-SD	99.15	94.20	99.18	95.36	97.42	68.77	3.24	12.7	91.42
YOLOv8n-SDB	99.15	94.79	99.10	95.02	97.44	69.07	3.24	12.7	89.03
YOLOv8n-SDBW	99.38	94.93	99.68	95.71	97.70	69.13	3.24	12.7	90.79

and the highest detection accuracy for Oil defects (99.28%), with an mAP@50 of 96.87% and an mAP@50:95 of 67.04%. In order to enhance the model's capacity for feature fusion and generalisation, the CBAM attention mechanism has been integrated into the SPPF module. This approach has been shown to be effective in improving the detection precision and recall of Scr and Sta defects, with mAP@50 and mAP@50:95 improving by 0.13% and 1%, respectively, without compromising detection speed. However, the enhanced algorithm still exhibits low detection accuracy for small-sized and challenging-to-distinguish defects and is susceptible to missing detections. To address these limitations, a small target detection layer is incorporated to enhance the detection accuracy for small-sized defects, as illustrated in Table 1. The YOLOv8-SD algorithm enhances the detection accuracy of Oil and Sta. The YOLOv8-SD algorithm demonstrates superiority over the YOLOv8-S algorithm in terms of mAP@50 and mAP@50:95, with enhancements of 0.42% and 0.73%, respectively. Additionally, the recall of the YOLOv8-SD algorithm is augmented by 1.25%, thereby effectively mitigating defect leakage. As some Scr defects exhibit minimal contrast, thus hindering their detection, the BiFPN module is integrated into the Concat module. This enhancement has been shown to elevate the detection accuracy of Scr defects by 0.59% in comparison to the YOLOv8-SD algorithm. In addition, marginal advancements have been observed in both mAP@50 and mAP@50:95. Finally, to address the issue of the Ciou loss function's inability to dynamically focus in the baseline algorithm, this paper replaces the Ciou with the WIoU v3 loss function, which possesses a dynamic non-monotonic focusing mechanism. Experimental results have demonstrated that the WIoU v3 loss function enhances the precision and recall of defect detection for various types of defects. This enhancement is attributed to the WIoU v3's advantageous properties, including its capacity for more rational gradient optimisation and improved adaptability to targets of diverse sizes, locations and shapes. In comparison with the original YOLOv8n algorithm, the YOLOv8n-SDBW algorithm, without a significant increase in model complexity, enhances detection precision by 1.06%, recall by 1.67%, mAP@50 and mAP@50:95 by 0.83% and 2.09%, respectively, and FPS reaches 90.79 frames per second, which meets the requirements for real-time detection of screen defects on mobile phones.

4.5. Comparison experiment

In order to evaluate the detection performance of the improved model, this study employs a comparative approach, contrasting the model under investigation with a number of widely used target detection models. All experiments are conducted on the same dataset and in the same experimental environment, and all algorithms are trained from scratch with a batch size of 32 and 200 iterations. The selected models comprise one-stage anchor-based approaches, including YOLOv8s, YOLOv8 m, YOLOv5s, YOLOv6s, EfficientNet, and two-stage anchor-based approaches, including Faster R-CNN.

The results of the comparison experiments are presented in Table 2. The enhanced network attains 97.99% precision, 95.71% recall, 97.7% mAP@50 and 69.13% mAP@50:95 for mobile phone screen defect detection, signifying a substantial enhancement in comparison with the original YOLOv8n algorithm. Furthermore, the YOLOv8-SDBW algorithm demonstrates superior precision and mAP@50 performance,

Table 2
Results of comparison experiment.

Model	P/%	R/%	mAP@50/%	mAP@50:95/%	Params	GFLOPs	FPS
YOLOv8n	96.93	94.04	96.87	67.04	3.01	8.1	113.64
YOLOv8s	97.37	95.43	97.50	69.42	11.13	28.4	57.19
YOLOv8m	97.17	95.69	97.58	71.48	25.84	78.7	22.27
YOLOv5s	97.32	95.28	97.34	69.11	9.11	23.8	52.91
YOLOv6s	96.95	93.78	96.32	67.78	16.3	44	40.98
EfficientNet	95.76	94.11	96.12	65.79	18.38	81.5	76.45
Faster R-CNN	95.32	94.48	95.38	64.32	41.36	179	10.27
YOLOv8n-SDBW	97.99	95.71	97.70	69.13	3.24	12.7	90.79

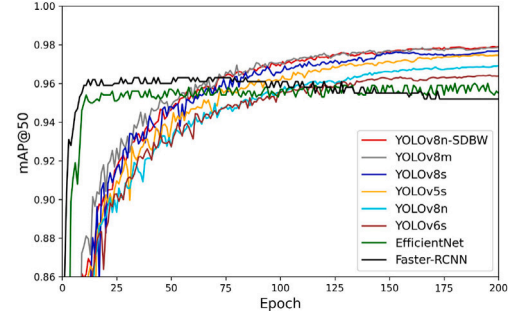


Fig. 8. Comparison of mAP@50 for the training process of each algorithm.

with 0.62% and 0.82% higher precision and 0.2% and 0.12% higher mAP@50, respectively, in comparison to YOLOv8s and YOLOv8 m. These enhanced algorithms exhibit superior recall, reduced parameter counts and model complexity, and accelerated detection speed. A comparison of the YOLOv8n-SDBW algorithm with YOLOv5s and YOLOv6s reveals superior metrics, with the YOLOv8n-SDBW algorithm exhibiting faster inference and higher detection efficiency. In addition, the YOLOv8n-SDBW algorithm significantly outperforms the EfficientNet and Faster R-CNN algorithms in terms of detection accuracy and efficiency. Furthermore, the number of parameters and model complexity are smaller, which makes it more suitable for the task of detecting mobile phone screen defects in real time.

In order to observe the detection performance of each model more intuitively, the mAP@50 of the training process of each model is plotted into one graph. As demonstrated in Fig. 8, the curves of the YOLOv8 m algorithm and the enhanced model are more closely aligned. However, it should be noted that the YOLOv8 m algorithm requires approximately eight times more parameters than the enhanced model, and the model complexity that is more than six times greater. Furthermore, the FPS of the YOLOv8 m algorithm is only a quarter of that of the enhanced model. In comparison with YOLOv8s, YOLOv5s, YOLOv8n, and YOLOv6s, the curves of the YOLOv8n-SDBW algorithm are located above them, indicating that the model detection is superior. In comparison with EfficientNet and Faster R-CNN, although the YOLOv8n-SDBW algorithm requires more training sessions to converge, the improved algorithm has a faster overall training speed and higher detection accuracy and efficiency.

In order to demonstrate the enhanced precision of the model, we have selected images from the dataset for comparison experiments in the context of detection to conduct a comparative analysis of the detection capabilities of the network before and after the implementation of the YOLOv8n enhancement. The results of the detection are presented in Fig. 9. A comparison of the detection results on both sides of Fig. 9 reveals that the enhanced model has led to an improvement in the accuracy of oil, scratches and stains defect detection. Furthermore, the enhanced model is capable of detecting minute defects and those with low contrast that were previously undetectable by the original model, as illustrated in Fig. 9(b1) and Fig. 9(b2). The experimental results demonstrate that the enhanced YOLOv8n model outperforms the initial model, reducing the leakage detection rate. This makes it suitable for use in mobile phone screen surface defect detection tasks, offering a valuable contribution to practical applications.

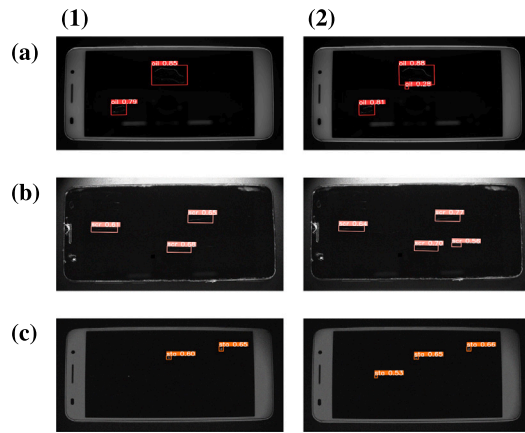


Fig. 9. Comparison of test results between YOLOv8n and the improved model of YOLOv8n; (1) YOLOv8n; (2) YOLOv8n-SDBW.

5. Conclusion

In this paper, an enhanced network architecture is proposed, based on YOLOv8n, for the efficient and precise detection of defects in mobile phone screens. The proposed network incorporates the CBAM attention mechanism into the SPPF module, facilitating multi-scale feature fusion capabilities within the network, thereby enhancing the detection accuracy of screen defects. The incorporation of a small target detection layer serves to enhance the detection accuracy of defects that are small in size and have low contrast, whilst concomitantly effecting a significant reduction in detection leakage. The BIFPN module is incorporated in the Neck part, thereby enhancing the model's ability to capture feature information without increasing the number of model parameters and model complexity. This serves to increase the detection accuracy of Scr defects that are characterised by ambiguity. Finally, this paper replaces the CIoU loss function of the original model with the WIoU v3 loss function. The purpose of this replacement is to reduce the harmful gradient generated by the low-quality examples, to make the model training more stable, and to improve the detection effect of the model without affecting the model detection speed. As demonstrated by the ablation and comparison experiments, in comparison with the initial YOLOv8n model, the number of parameters of the YOLOv8n-SDBW model increases by a mere 0.24M, and the model complexity increases by 4.6G. Concurrently, the precision rate, recall rate, mAP@50, and mAP@50:95 have been enhanced by 1.06%, 1.67%, 0.83%, and 2.09%, respectively. In comparison with algorithms such as YOLOv8s and YOLOv8 m, which demonstrate superior detection performance, the enhanced model exhibits higher detection accuracy, reduced leakage rate, accelerated detection speed, and does not necessitate excessive computational resources. Consequently, the YOLOv8n-SDBW algorithm proposed in this paper can effectively and accurately detect screen defects on mobile phones without requiring excessive computational resources and storage space.

CRedit authorship contribution statement

Peng Shi: Writing – original draft. **Xueqin Li:** Writing – review & editing. **Zhiming Feng:** Conceptualization. **Xingwang Shang:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] M.T. Islam, A.B. Abdullah, S. Shahir, M.A. Kalam, H.H. Masjuki, R. Shumon, M.H. Rashid, A public survey on knowledge, awareness, attitude and willingness to pay for WEEE management: Case study in Bangladesh, *J. Clean. Prod.* 137 (2016) 728–740.
- [2] V. Forti, C.P. Balde, R. Kuehr, G. Bel, The Global E-waste Monitor 2020: Quantities, Flows and the Circular Economy Potential, United Nations University/United Nations Institute for Training and Research ..., 2020.
- [3] S. Zhang, Y. Ding, B. Liu, C.-c. Chang, Supply and demand of some critical metals and present status of their recycling in WEEE, *Waste Manage.* 65 (2017) 113–127.
- [4] B. Tansel, From electronic consumer products to e-wastes: Global outlook, waste quantities, recycling challenges, *Environ. Int.* 98 (2017) 35–45.
- [5] M. Kaya, Recovery of metals and nonmetals from electronic waste by physical and chemical recycling processes, *Waste Manage.* 57 (2016) 64–90.
- [6] F. Cucchella, I. D'Adamo, S.L. Koh, P. Rosa, Recycling of WEEEs: An economic assessment of present and future e-waste streams, *Renew. Sustain. Energy Rev.* 51 (2015) 263–272.
- [7] L. Lou, K. Lu, J. Xue, Defect detection based on improved YOLOx for ultrasonic images, *Sens. Imaging* 25 (1) (2024) 10.
- [8] S. Hou, B. Dong, H. Wang, G. Wu, Inspection of surface defects on stay cables using a robot and transfer learning, *Autom. Constr.* 119 (2020) 103382.
- [9] S.-H. Chen, Y.-W. Lai, C.-L. Kuo, C.-Y. Lo, Y.-S. Lin, Y.-R. Lin, C.-H. Kang, C.-C. Tsai, A surface defect detection system for golden diamond pineapple based on CycleGAN and YOLOv4, *J. King Saud University-Comput. Inf. Sci.* 34 (10) (2022) 8041–8053.
- [10] B. Bosquet, D. Cores, L. Seidenari, V.M. Brea, M. Mucientes, A. Del Bimbo, A full data augmentation pipeline for small object detection based on generative adversarial networks, *Pattern Recognit.* 133 (2023) 108998.
- [11] W. Zhou, R. Zhou, Vision SLAM algorithm for wheeled robots integrating multiple sensors, *Plos One* 19 (3) (2024) e0301189.
- [12] H. Lou, X. Duan, J. Guo, H. Liu, J. Gu, L. Bi, H. Chen, DC-YOLOv8: small-size object detection algorithm based on camera sensor, *Electronics* 12 (10) (2023) 2323.
- [13] Y. He, K. Song, Q. Meng, Y. Yan, An end-to-end steel surface defect detection approach via fusing multiple hierarchical features, *IEEE Trans. Instrum. Meas.* 69 (4) (2019) 1493–1504.
- [14] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, et al., Searching for mobilenetv3, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019, pp. 1314–1324.
- [15] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, Ssd: Single shot multibox detector, in: *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, the Netherlands, October 11–14, 2016, Proceedings, Part I 14*, Springer, 2016, pp. 21–37.
- [16] S. Ren, K. He, R. Girshick, J. Sun, Faster r-cnn: Towards real-time object detection with region proposal networks, *Adv. Neural Inf. Process. Syst.* 28 (2015).
- [17] M. Hussain, YOLO-v1 to YOLO-v8, the rise of YOLO and its complementary nature toward digital manufacturing and industrial defect detection, *Machines* 11 (7) (2023) 677.
- [18] B. Mahaur, K. Mishra, Small-object detection based on YOLOv5 in autonomous driving systems, *Pattern Recognit. Lett.* 168 (2023) 115–122.
- [19] Z.-C. Yuan, Z.-T. Zhang, H. Su, L. Zhang, F. Shen, F. Zhang, Vision-based defect detection for mobile phone cover glass using deep neural networks, *Int. J. Precis. Eng. Manuf.* 19 (2018) 801–810.
- [20] Y. Xia, Y. Lu, X. Jiang, M. Xu, Enhanced multiscale attentional feature fusion model for defect detection on steel surfaces, *Pattern Recognit. Lett.* 188 (2025) 15–21.
- [21] Y. Zhu, R. Ding, W. Huang, P. Wei, G. Yang, Y. Wang, HMFA-Net: Hierarchical multi-frequency based Channel attention net for mobile phone surface defect detection, *Pattern Recognit. Lett.* 153 (2022) 118–125.
- [22] H. Li, P. Xiong, J. An, L. Wang, Pyramid attention network for semantic segmentation, 2018, arXiv preprint arXiv:1805.10180.
- [23] C. Feng, Y. Zhong, Y. Gao, M.R. Scott, W. Huang, Toood: Task-aligned one-stage object detection, in: *2021 IEEE/CVF International Conference on Computer Vision, ICCV*, IEEE Computer Society, 2021, pp. 3490–3499.
- [24] S. Woo, J. Park, J.-Y. Lee, I.S. Kweon, Cbam: Convolutional block attention module, in: *Proceedings of the European Conference on Computer Vision, ECCV*, 2018, pp. 3–19.
- [25] M. Tan, R. Pang, Q.V. Le, Efficientdet: Scalable and efficient object detection, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 10781–10790.

- [26] Q. Cai, Y. Pan, Y. Wang, J. Liu, T. Yao, T. Mei, Learning a unified sample weighting network for object detection, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 14173–14182.
- [27] Z. Tong, Y. Chen, Z. Xu, R. Yu, Wise-IoU: bounding box regression loss with dynamic focusing mechanism, 2023, arXiv preprint [arXiv:2301.10051](https://arxiv.org/abs/2301.10051).
- [28] Y. Ding, C. Liu, H. Zhu, Q. Chen, A supervised data augmentation strategy based on random combinations of key features, Inform. Sci. 632 (2023) 678–697.